

Computer System

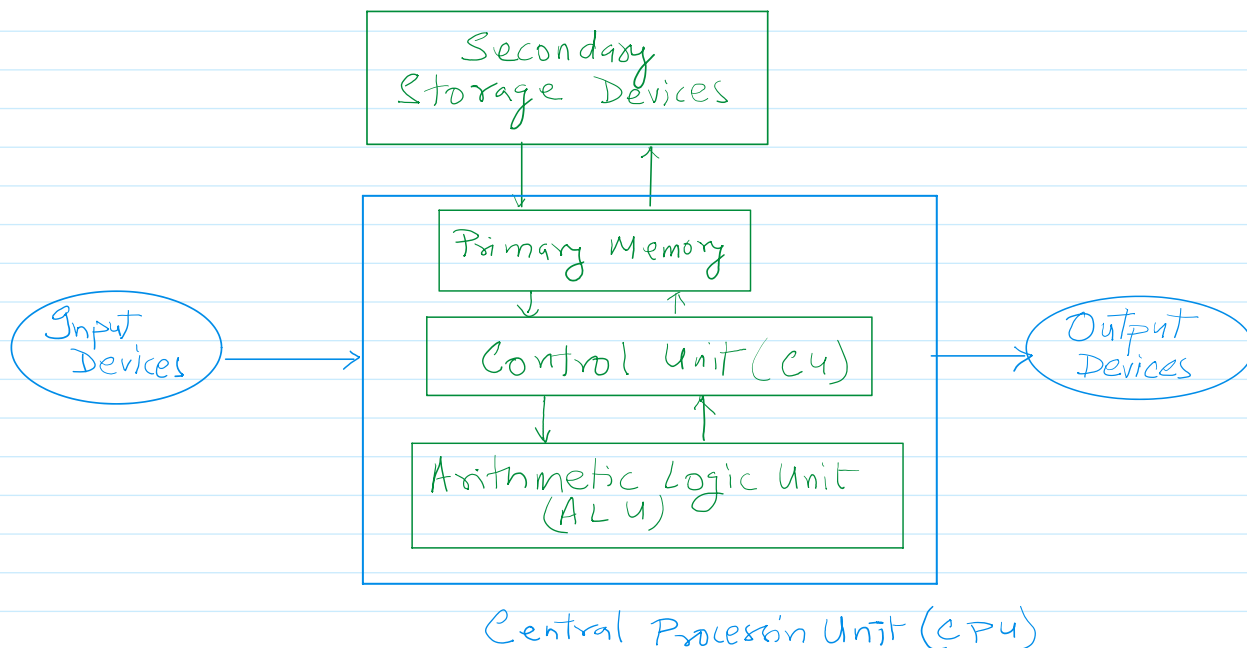
26 July 2026 09:38

A computer is an electronic device that can be programmed to accept data (input) process it, and generate a result (output).

What is computer system?

A computer along with additional hardware and software together is called a **computer system**.

A computer system primarily comprises a central processing unit (CPU) memory, Input/output devices and storage devices. All these components function together as a single unit to deliver the desired output. A computer system comes in various forms and sizes. It can vary from high end server to personal desktop.



Central Processing Unit(CPU)

It is the electronic circuitry of a computer that carries out the actual processing and usually referred as the brain of the computer. It is commonly called processor also. Physically, a CPU can be placed on one or more microchips called integrated circuits (IC). The ICs comprise semiconductor materials.

The CPU is given instruction and data through programs. The CPU then fetches the program and data from the memory and performs arithmetic and logic operations as per the given instructions and store the result back to memory.

While processing the CPU stores the data as well as instructions in its local memory called registers. Registers are part of the CPU chips and they are limited in size and number. Different registers are used for storing data instructions or intermediate results.

Other than registers, the CPU has two main components - Arithmetic logic unit (ALU) and control unit (CU). ALU performs all the arithmetic and logic operations that need to be done as per the instruction in a program. Control unit controls sequential instruction execution, interprets instructions, and guides data flow through the computer's memory ALU and input or output devices. CPU is also popularly known as microprocessors.



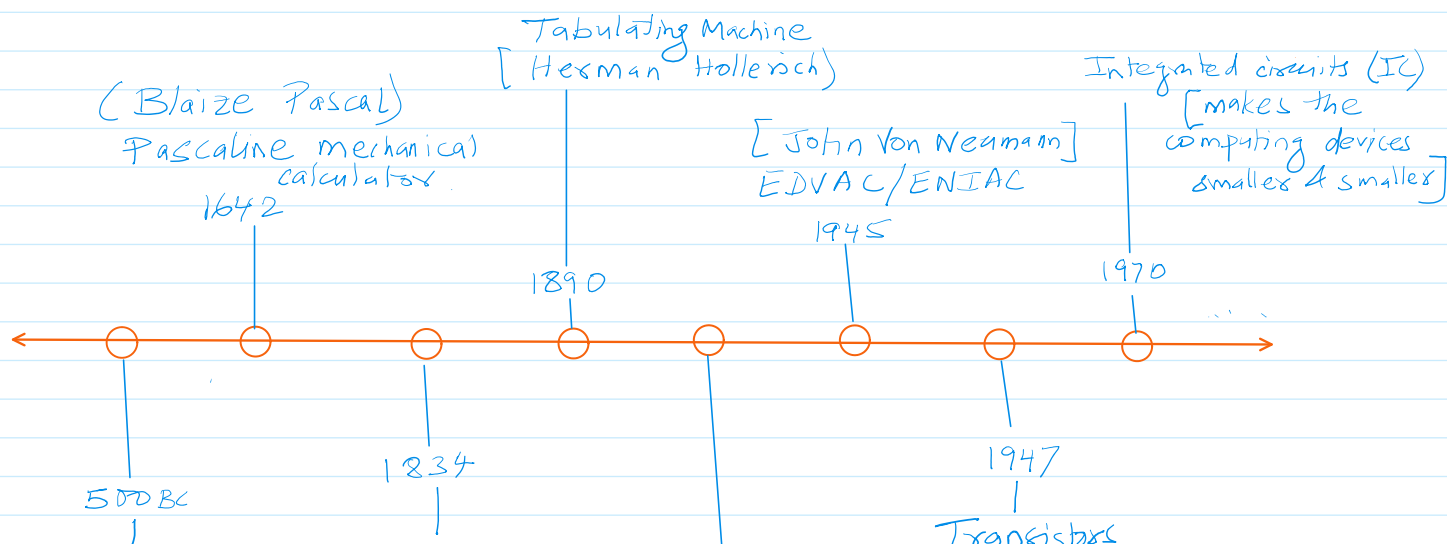
Input Devices: The devices through which control signals are sent to the computer are termed as input devices. These devices convert the input data into a digital form that is acceptable by the computer system. Some examples of input devices include keyboard, mouse, scanner, touch screen, etc.

Data entered through input devices is temporarily stored in main memory (also called RAM) of the computer system. For permanent storage and future use, the data as well as instructions are stored permanently in additional storage location called secondary memory.

Output devices: The device that receives data from a computer system for display, physical production, etc. Is called output devices. It converts digital information into human understandable form. For example, monitor, projector, headphones, speaker, printer, etc.

Evolution of computer

From the simple calculator to a modern day powerful data processor, computing devices have evolved in relatively short span of time. The evolution of computing devices is shown through a timeline. In the following figure:



500 BC
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Abacus

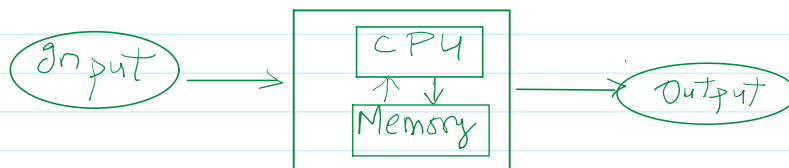
1834
|
Analytical Engine
(Charles Babbage)

1947
|
Transistors
[vacuum Tubes were replaced by transistors developed at Bell Labs, using semiconductor materials]

1937
|
Turing Machine

[concept was a general purpose programmable machine that was capable of solving any problem by executing the program stored on the punched cards.]

Timeline showing key inventions in computing technology.



Von Neumann architecture for the computer

ENIAC [Electronic Numerical Integrated And Computer]